

WILLIAM EWANCHUK

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EDUCATION

University of Alberta | B.Sc. Computer Engineering Co-op | Expected May 2027

EXPERIENCE

Full Stack Developer/DevOps

April 2026 – Current

Company: North American Construction Group

Stack - Azure DevOps, Static Web Apps, Azure Container Apps/KEDA, Docker/ACR, .NET 10, .NET Core, React/Tanstack, React Native, SQLite, OpenTelemetry, Cloudflare

- Developing the web application and .NET backend for a company-wide rewrite of the field maintenance platform, used across NACG subcontractors at North American and Australian mining sites for downtime reporting, parts requests, equipment restoration, and tech scheduling – replacing manual JDE Oracle coordination with a direct integration
- Containerized the full alpha deployment, caching package dependencies and pre-baking VM tooling images for CI pipelines, cutting build times by **80%**
- Owned the Azure Blob storage feature from design to production – defined backend access patterns using user-delegation SAS tokens across work orders, tech notes, and RNA alerts (image, video, and PDF), built temporary frontend upload paths with server-confirmed uploads, and wrote automated cleanup jobs for orphaned uploads
- Instrumented monitoring with **OpenTelemetry** and Application Insights, adding user-controlled log-level tracing for beta testers and custom telemetry filtering
- Designed and deployed the beta release pipeline end-to-end – database migrations, changelog generation, ACR image builds, ACA revision deployments, log aggregation, and blue/green traffic switching with zero-downtime health probing across staging and production
- Built offline mode for the React Native app using a SQLite-backed query queue – requests made without connectivity persist locally and replay on reconnect, so field workers can create and update work orders without service

Software Engineering Intern

November 2024 – August 2025 (10 months)

Company: Stepscale

Stack - Django, PostgreSQL, LangGraph, LangChain, OpenRouter, AWS Elastic Beanstalk, CloudWatch, Docker, Pipedream, Stripe, Amplitude, Redis, GitHub Actions

- Built an isolated Docker sandbox for LangGraph agents to run Python file generation and analysis with automated output validation, improving safety and reproducibility
- Refactored the report export pipeline to support PDF, Word, Excel, and Google Workspace output formats
- Implemented a **Redis** caching layer to pre-warm LangChain conversation chains for connected integrations, reducing redundant inference requests and improving response latency by **40%**
- Integrated Amplitude analytics to track tool call errors, navigation, and landing page bounce rates, identifying failing flows and high drop-off pages
- Designed and secured backend endpoints for **Stripe** billing, including pricing tiers, subscriptions, and webhooks
- Monitored and debugged deployments with AWS CloudWatch and GitHub Actions across dev and prod environments

PROJECTS

LittleBrick 3D Printing – E-Commerce Platform

GitHub | Live Demo

Stack - Spring Boot, PostgreSQL, Next.js, TypeScript, Three.js, React Three Fiber, Stripe, Shippo, Google Maps, Resend, WebSocket/STOMP, Docker Compose, Nginx, Terraform, Ansible, Tailscale, Cloudflare Tunnels, GitHub Actions

- Built an interactive 3D product preview using Three.js and React Three Fiber – loads multi-part STL meshes into a Rapier physics simulation with keyboard movement, pointer-lock dragging, and orbit camera controls for inspecting prints before ordering
- Integrated Google Maps to show live shipment location and destination markers on the order page, geocoding addresses server-side via the Google Maps API and caching coordinates on the shipping record
- Estimated parcel dimensions server-side from cart item footprints and weights to generate live Shippo rate quotes at checkout; admin purchases the final label with actual box dimensions via Shippo's Transaction API
- Coordinated Stripe and Shippo webhooks to drive the order lifecycle – Stripe events advance order status, Shippo

tracking events update the shipping record and trigger geocoding in a single write

- Replaced Spring Security's default session handling with PostgreSQL-backed bearer sessions, implementing lazy user attachment and role-based endpoint authorization
- Provisioned the production server on Proxmox with Terraform and Ansible – Tailscale registration, Stripe CLI as a systemd webhook forwarder, Docker Compose stack, and self-hosted GitHub Actions runners on the LAN with NFS-backed state

EKS Traffic Driven Scaling Platform: DevOps/SRE

GitHub | Live Demo

Stack - Kubernetes, Helm, EKS, KEDA, Grafana, Prometheus, Terraform, Python, Docker, Redis, AWS CLI, GitLab

- Deployed a FastAPI service to AWS EKS using Terraform, with automated TLS, monitoring, and PromQL autoscaling to benchmark Gunicorn worker and thread configurations
- Automated a CI/CD style deployment pipeline for AWS EKS using AWS CLI and Helm, including Route 53 DNS updates and NGINX Ingress TLS certificate provisioning with Let's Encrypt
- Enabled traffic based autoscaling with KEDA and Prometheus metrics, monitored in Grafana and load tested with k6, sustaining **5k+ RPS at 500 VUs**
- Built a lightweight dashboard for internal pod metrics and optimized CPU metric collection with a Redis shared cache to avoid redundant per container polling

SKILLS AND AWARDS

Languages - Python, JavaScript, TypeScript, Java, C#, HTML, CSS, Bash, SQL, C, C++

Frameworks - .NET, React.js, React Native, Next.js, Nuxt.js, Django, Express, Spring Boot, FastAPI, Three.js

Databases - PostgreSQL, MySQL, SQLite, MongoDB

Cloud and DevOps - Kubernetes, KEDA, Docker, Azure (Container Apps/ACR/Static Web Apps/Blob Storage), AWS (EKS/EC2/S3/RDS/CloudWatch), Cloudflare, Tailscale, Nginx, Terraform, Ansible, Helm, Linux/Unix

Dev Tools and Observability - Git, GitHub Actions, GitLab CI, Azure DevOps, Grafana, Prometheus, OpenTelemetry, Application Insights, CloudWatch, JUnit, Figma